

Data Science Club – Football Analyses

Project Overview

Hungarian football currently lacks strong competition in the field of performance analytics, creating an opportunity to build a modern, scalable analytics platform. The aim of this project is to develop an application that extracts football insights from freely available data and match videos, turning them into actionable information for clubs.

The system will analyze matches and individual players to reveal patterns that support training decisions, match preparation, and player recruitment. Each club would receive team-specific insights, with the long-term goal of working together to unlock more value from their internal data.

The project started with the intention of supporting Győri ETO as a proof of concept, but the platform is designed to scale to any Hungarian club interested. The ultimate objective is to uncover the underlying information within matches and player performance, helping teams gain a competitive advantage.

Team & Responsibilities

Team Member	Role	Key Responsibilities	Informarion Source
Géza Mihályi	Project Lead	Planning, coordination, progress tracking, Forecast, Simulations	link1 link2 link3 link4 link5 link6
Member 2	Data Engineer	Data collection, cleaning, preprocessing, Data Validation	link1 link2 link3 link4 Also Videos from the Hungarian League
Member 3	Data Engineer	Same	same
Member 4	Data Scientist	Statistical models, Possession simulations	Data Scientist will main works in the following chapters from the page: link

			Chapter 2, Chapter 5, Chapter 8, Chapter 6
Member 5	Data Scientist	Markov Simulation	It helps to see what are the chances from a given position that the team will score. It is written pretty well in the „How to win the Premier League book“. I will help here to explain and create the simulation.
Member 6	Machine Learning Engineer	Valuing Actions, xG, PSxG, Expected Threat	Machine Learning Engineers will mainly work in the following chapters from the page: link Chapter 4, Chapter 9. link
Member 7	Machine Learning Engineer	Video Analyses, Object detection from videos, Heat Map, Situation sorting.	link link It will be discussed what we plan to achieve from this.
Member 8	Analyst	Insights, Solution proposition	Some Football Knowledge
Member 9	Visualization	Dashboards, plots, storytelling	link1

There are some books that are pretty good starters for this, I will include their link. They are downloadable from pirate sites (But I think if you are interested in football and data science you should get a copy.)

[Soccermetrics](#)

[Football Hackers](#)

[How to win the Premier League](#)

If you have any good book about Football and Data analytics, feel free to share

? Problem Statement

Hungarian football clubs have limited access to affordable, high-quality performance analytics tools that can turn match data and video into practical, decision-support insights. While elite international teams increasingly rely on data-driven analysis for training, match preparation, and recruitment, most domestic clubs lack the infrastructure and expertise to extract meaningful value from available data sources.

As a result, valuable information hidden in match footage and publicly available datasets remains unused, creating a competitive gap. There is a need for a scalable analytics solution that can transform raw football data into clear, team-specific insights, enabling Hungarian clubs to make more informed sporting and strategic decisions.

Data Description

- **Data Sources:** Public Datasets, APIs, web scraping, simulated data
- **Data Types:** Tabular, Video, Time-Series
- **Size & Limitations:** Each Team plays 33 matches and there are 12 matches, Each match has some number of data, so the data size maximum in the thousands, not in the hundred thousands. Match lengths are around 1:30 hours. When we extract exact data from the matches, then the match data could be in the 100,000 sizes. There are no publicly available, dataset for Hungarian league that contains the position of players, or the actions by sub-minute granularity. In the beginning we will use publicly available benchmark datasets to setup the pipeline, and after making our own data preparation from match videos we can start to use ours.

Key Challenges

- Limited data on Hungarian matches → we need to collect, and start a big Database (Which we could potentially, sell)
- Difficult and Tedious Data Engineering → Get the sub-minute data from a football match

Methodology (High Level)

1. Data Collection (Team Member 2 and 3)
2. EDA and Feature Engineering. (Team Member 2 and 3)

Here it goes to 4 ways:

3. Model building
 - a. xG models, and Valuing Actions (Team Member 7)
 - b. Video Analyses, Heatmaps and Situation Sorting (The Sorted situations can be used in other statistical analyses procedures, or in the xG models and Valuing actions. (Team Member 6)
 - c. Statistical models, Randomness and predictions (Team Member 4)
 - d. Pitch Control, and player pitch coverage (Team Member 5)
4. Validation → Check if the models found to be useful (Team Member 8)
5. Result interpretation (Team Member 8)
6. Visualization (Team Member 9)

Project Deliverables

- Plots about given metrics like xG, PSxG, passing networks
- Simulations of a given matches, given leagues. Goal distribution.
- Result interpretation PDFs

Tools & Technologies

- **Languages:** Python, SQL
- **Data Processing:** Pandas, NumPy,
- **Machine Learning:** Scikit-learn, XGBoost, PyTorch
- **Visualization:** Matplotlib, Seaborn
- **Storage:** SQL databases
- **Collaboration:** Git, GitHub
- **Anything else that is new is welcome**, I would like to try new technologies

Timeline & Milestones

- **Week 1:** Gather data, everyone tries to help in gathering data, watching footage from the past matches that already happened in the club season.
- **Week 2:** Now we collected some data about a match, create an SQL server, that every Team member can reach. (I propose that it will be in the in my raspberry pi, that we can all connect to via tailscale). The SQL server will hold every match data, so we have to figure out how to setup the SQL tables, what goes in, what will be the master etc. Create a Pipeline that uploads the data collected.
- **Week 3:** From now on, the Data Scientist Members and the Machine Learning members can start turning the data into something useful, based on the available resources.
- **Week 4:** Continue the data collecting and model building

- **Week 5:** Here I would like to see results in the Passing Network (Data Scientist Team), and in the xG models (ML Team).
- **Week 6:** Markov Simulation
- **Week 7:** By that, I would to have a pipeline, where we can select a given match that has already happened and create plots based on the Passing Networks, xG models and the Markov simulation
- **Week 8:** The first written report from the data. And also Whatever simulations and action where we have time.
- **Week 9: Demo day. Show everything that we created. Decide if this is worth trying as a company, and if yes, who would like to be a part of it.**

From this it can be seen that the main scopes that I would like to achieve is a passing network, Markov Simulations and xG model. Anything else we manage to do is plus. While we do not have data from the match, the ML and DS team can scrape the web for information to start the simulations and models.

Final Results & Outcomes

- Final outputs and deliverables.
- Results.
- Feedback. Really important, honest feedback to each other, to the Team Leader from the Team, to The Team from the Team Leader.
- Other Limitations that we found.
- **My ultimate goal is to find a team, with whom we can start a company first in part-time that would create analytics for the Hungarian clubs.**
- **It is not a problem if we did not do everything we planned. The more the better, obviously, but to find a team that could work together in the future also.**